TEXORA · LECTURE SHEET

Jute Processing and Batching

Jute arrives not as single fibres but as long, stiff, gummed reeds. Batching is where the mill decides what it will spin — mixing the grades to a cost and quality target, and softening the batch with oil and water so the reeds survive carding.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 5 · LEVEL 2 TERM 1

1. The jute fibre the mill receives

Jute is a **bast fibre**, stripped from the stem of *Corchorus* plants (white jute, *C. capsularis*; tossa jute, *C. olitorius*) after **retting** — soaking the stems so microbial action loosens the fibre bark from the woody stick. What the mill buys is not an ultimate fibre but a **reed**: a ribbon of bast, commonly 1.5–3.5 m long, of many short **ultimate fibres** (each about 1.5–5 mm long, 15–25 µm across) cemented together by natural gums — chiefly **pectins, hemicelluloses and lignin**. The reed is strong in tension but stiff, harsh and brittle when dry, and its fibres cannot slide over one another as cotton fibres do. Almost everything in jute processing follows: the spinning unit is a long reed, its cement must be softened before drafting, and the material is far too long and entangled for a cotton-type card.

The reed is judged on **strength** (it must survive breaker-card pinning without breaking short), **length** (long reeds draft better, with fewer ends-down), **colour and lustre** (bright, golden fibre commands the better yarns), **fineness and softness** (finer, well-retted fibre spins finer counts) and **freedom from defects**. Moisture also matters, because jute is traded by weight: normal regain is about 12–14 %; over-dry fibre is brittle, over-wet fibre heats in store and mildews.

2. The jute mill route in outline

Each stage solves one part of the reed problem: soften it, open it, parallelise it, draft it, twist it.

1. **Batching** — bales are opened and the reeds made up into **batches**: deliberate mixtures of grades chosen for the yarn count, quality and cost required (this topic).
2. **Softening** — the batch is sprayed with batching emulsion and piled for 24–48 h so the gums soften (this topic).
3. **Spreading** — the softener/spreader carries the reeds on a lattice under spray and delivers a continuous sheet of overlapping reeds, evening the feed for the card.

4. **Breaker card** — pinned rollers tear the reeds open and break the network down toward its ultimate fibres, delivering a heavy sliver (Topic 6).
5. **Finisher card** — finer, denser pinning repeats the opening, delivering a more uniform sliver (Topic 6).
6. **Drawing frames** — two or three passages of pinned (gill-type) drawing draft and double the slivers toward spinning weight.
7. **Spinning** — flyer frames draft further and insert twist. Jute counts are coarse and expressed in the jute system (pounds per spindle of 14,400 yards).
8. **Winding and reeling** — yarn is wound into packages for weaving, or reeled into hanks (**reeling**) for sale; woven, it becomes hessian, sacking and carpet backing.

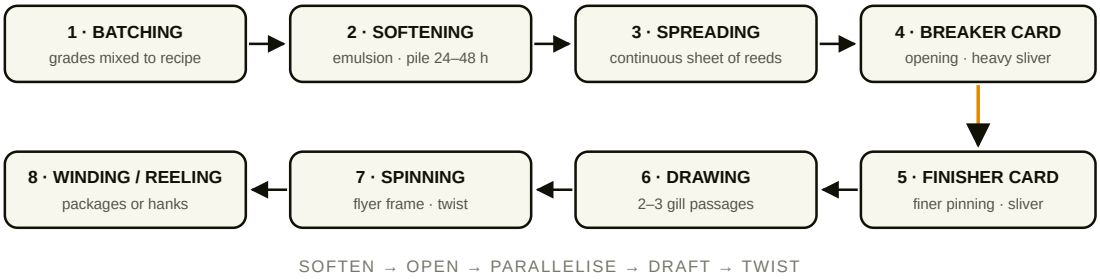


Figure 1 — The jute mill route. Batching fixes the quality and cost of everything downstream.

3. Jute grades

No two agricultural lots are identical, so the trade sorts fibre into **grades** by type and quality — strength, length, colour, fineness and freedom from defects. Tossa jutes are generally stronger, finer and more lustrous than white jutes and take the better grades; within each type the scale runs from top grades (long, strong, bright, clean) down to low grades (short, weak, dull, defective). At the bottom sit **cuttings** — the hard root ends — sold separately and used only in coarse batches. The batcher selects not individual reeds but proportions of named grades whose card-room behaviour is known.

GRADING FACTOR	WHAT IS JUDGED	EFFECT IF POOR
Strength	Resistance to breaking under tension and pinning	Breaks short in carding; weak yarn; ends-down
Length	Reed length, typically 1.5–3.5 m	Short reeds draft badly; more waste, irregular yarn
Colour and lustre	Brightness — white to golden to reddish, by type and retting	Dull fibre restricts the batch to dyed or coarse goods

Fineness and softness	Reed fineness and pliability after retting	Coarse, harsh fibre will not spin fine counts
Defects	Bark, specks, roots, stick and stain (see §4)	Survive into the yarn as faults and weak places

4. Defects the batcher must control

- **Bark (barky jute)** — stem bark still adhering because retting or stripping was incomplete. Bark does not soften or separate in carding; it passes forward as hard, dark flakes visible in yarn and cloth.
- **Specks** — small dark spots and adhering fragments scattered along the reed. A specky lot downgrades the whole batch, because specks distribute through every sliver it produces.
- **Roots (root content)** — the hard, woody butt end of the reed, cut off as *cuttings* before batching for better yarns; uncut roots break into short, harsh pieces that will not draft and overload the card pins.
- **Sticks and hurd** — pieces of woody stem core; largely removed in carding as waste, but they damage pins and break fibres on the way.
- **Stain and mildew** — from careless retting or wet storage; stained fibre is weak and unevenly coloured and must be kept out of light-coloured goods.
- **Over- or under-retted fibre** — over-retting rots the fibre itself (weak, dull); under-retting leaves the gums hard (harsh, barky, difficult to card).

The batcher's defence is dilution and exclusion: heavily defective lots are rejected, downgraded to a coarse batch, or cleaned by cutting. Batching can never remove a defect — only decide how much the yarn must tolerate.

5. Batch selection — making a batch to a recipe

A **batch** is the quantity of fibre — several grades, often from several bales and lots — processed together as one blend for a stated purpose. Batch selection balances three targets that pull against each other:

- **The yarn count.** Finer counts demand longer, stronger, finer, cleaner fibre — the strand is thinner and any weak element shows at once. Coarse sacking counts tolerate short, lower-grade fibre and cuttings.
- **The yarn quality.** End use sets the floor: a hessian warp yarn (sized, tensioned, seen) needs a better batch than a weft yarn; undyed yarn needs colour discipline a dyed sacking yarn does not.

- **The cost.** The batcher meets the specification with the *cheapest mixture that will still spin it*: a backbone of sound medium grade, better grade to lift strength and appearance, and — for coarse work — cheap grade or cuttings to carry cost down, never so much that spinning collapses.

In practice the batcher works from standard **batch recipes** — proven percentage mixtures for each regular yarn — adjusted to the stock in hand and its prices. Fibre that is too good wastes money on every kilogram spun; fibre that is too poor wastes far more, in card waste, ends-down, weak yarn and rejected cloth. Once made, the batch is kept together through softening and carding so its properties stay uniform for the whole order.

6. The batching emulsion

6.1 Why jute must be softened

Dry reed is held rigid by its pectins and gums. Fed dry to the breaker card it would snap rather than separate: length destroyed, waste and pin damage soaring. **Softening** treats the batch with an **oil-in-water emulsion** and time: water swells and softens the gums while oil lubricates the surfaces. After conditioning the reeds are pliable — in the card the cement yields progressively, the network opens toward its ultimate fibres with far less breakage, and fibres slide in drafting instead of tearing.

6.2 Composition

The emulsion is a stable dispersion of fine oil droplets in water, held together by a small dose of **emulsifier** (a surface-active agent). A typical mill recipe is of the order of:

COMPONENT	TYPICAL SHARE	FUNCTION
Oil (batching oil — mineral or vegetable-based)	≈ 15–25 %	Lubricates fibre surfaces; reduces brittleness and friction in carding and drafting
Water	≈ 74–84 %	The carrier; swells and softens the pectins and gums; restores pliability
Emulsifier	≈ 1 %	Keeps the oil dispersed as fine droplets so every reed receives an even share

The emulsion is made up in a tank — water first, emulsifier dispersed, oil added slowly with agitation — and used promptly: a broken (separated) emulsion sprays unevenly and the batch softens patchily.

6.3 How much is applied — oil add-on

Application is quoted two ways, never to be confused. The **emulsion add-on** is the emulsion applied as a percentage of fibre weight — commonly 20–30 %. The **oil add-on** (oil on fibre) is only its oil

fraction: at 25 % add-on of a 20 %-oil emulsion, oil on fibre is 25 % × 20 % = 5 %. Too little oil leaves fibre harsh and breakage-prone; too much makes fibre cling to pins, loads the cards and wastes oil.

WORKED EXAMPLE 1 — EMULSION MAKE-UP

Prepare 100 L of batching emulsion containing 20 % oil, 79 % water and 1 % emulsifier (parts by volume/mass as batched in the tank).

Oil = 100 × 20 ÷ 100 = **20 L of oil**.

Water = 100 × 79 ÷ 100 = **79 L of water**.

Emulsifier = 100 × 1 ÷ 100 = **1 (L/kg) of emulsifier**.

Check: 20 + 79 + 1 = 100. Make up: emulsifier in the water first, oil added slowly under agitation.

WORKED EXAMPLE 2 — APPLICATION AND OIL ON FIBRE

A batch of 400 kg of jute is to receive an emulsion add-on of 25 % using the emulsion of Worked example 1 (20 % oil).

Emulsion required = 400 kg × 25 ÷ 100 = **100 kg of emulsion**.

Oil in that emulsion = 100 kg × 20 ÷ 100 = **20 kg of oil**.

Oil on fibre = 20 kg ÷ 400 kg × 100 = **5 % oil on the weight of fibre**.

The remaining 80 kg is essentially water (plus 1 kg emulsifier), which softens the gums and then partly evaporates.

7. Application, piling and conditioning

The emulsion is applied where the fibre is on the move, so it spreads evenly. In the **softener**, the reeds (laid as handfuls called **stricks**) pass between pairs of heavy fluted rollers under a spray; the rollers squeeze and crimp the reeds — the working itself helps break the reed's stiffness — while the spray wets every layer. In the **spreader**, a lattice carries the reeds under spray bars and delivers an overlapped sheet. Many mills combine both in a softener–spreader line. The essentials: meter the emulsion to the set add-on, spray across the full width, and work the reeds so no dry core survives inside a thick strick.

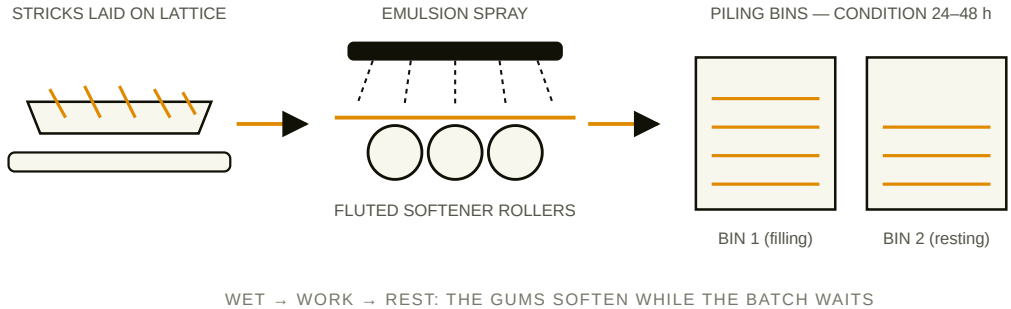


Figure 2 — Batching and softening: stricks sprayed through the softener rollers, then piled in bins to condition, first in first out.

Sprayed fibre is not yet softened fibre. It is stacked in **piling bins** and left to **condition — typically 24–48 hours**, while the emulsion migrates into the reed interiors and water swells the pectins: fibre taken too early cards harshly, with dry, brittle cores. Bins are worked first in, first out, so every part gets its full time; over-long piling is avoided, since a wet, oily mass can heat, ferment and stain.

8. Purposes of the emulsion — summary

The emulsion exists to **soften the gums and pectins** (the water's work, given piling time, so the network opens instead of snapping), to **lubricate** fibre and metal contact (the oil's work, cutting pin breakage and letting fibres slide in drafting), to **reduce brittleness** by restoring moisture and pliability, and to **make processing uniform** — patchy softening is patchy carding, and no drawing passage fully repairs it.

Glossary

Batch — a deliberate mixture of grades processed together for a stated yarn. **Batching emulsion** — the oil-in-water spray (oil, water, emulsifier) used to soften the batch. **Strick** — a handful bundle of reeds, the hand-fed unit at the softener. **Piling / conditioning** — resting sprayed fibre in bins, typically 24–48 h, while the emulsion penetrates. **Emulsion add-on** — emulsion applied as a percentage of fibre weight. **Oil on fibre** — the oil fraction alone, as a percentage of fibre weight. **Reed** — the traded strand: ultimate fibres cemented by pectins and lignin. **Cuttings** — hard root ends cut from reeds, used only in coarse batches. **Bark** — stem bark still adhering to the fibre. **Specks** — small dark adhering fragments in a lot. **Spreader** — the lattice machine delivering the softened reeds as a continuous sheet.

Key points

• Jute is spun as a gummed reed of cemented ultimate fibres — softening must precede opening, or the fibre breaks instead of separating.
• Mill route: batching → softening → spreading → breaker card → finisher card → drawing → spinning → winding/reeling.
• A batch is a recipe: grades proportioned to meet count and quality at the lowest cost that will still spin.
• Defects — bark, specks, roots, sticks, stain — cannot be removed by batching, only excluded, cut out, or diluted.
• Emulsion is oil dispersed in water by an emulsifier; water softens the gums, oil lubricates and reduces brittleness.
• Emulsion add-on (≈ 20–30 %) and oil on fibre (add-on × oil fraction) differ — e.g. 25 % × 20 % = 5 % oil on fibre.

• Sprayed fibre must pile typically 24–48 h, first in first out; softening is finished in the bin, not at the spray bar.

Practice

1. Prepare 250 L of emulsion at 18 % oil, 81 % water and 1 % emulsifier. State each component quantity and check they sum to 250 L.
2. A 600 kg batch receives a 22 % add-on of a 20 %-oil emulsion. Calculate the emulsion applied, the oil applied, and the oil on fibre (%).
3. A mill must spin (a) a fine, undyed hessian warp yarn and (b) a coarse dyed sacking weft. How should the two batches differ in grades and defect tolerance, and why?
4. Explain, from the reed's structure, why jute cannot be carded dry as cotton is carded.
5. Harsh carding and high breakage come from one bin but not the next, from the same batch. Give three batching-side causes to check before blaming the card.

Answer key

1. Oil = 250 × 18 ÷ 100 = **45 L**; water = 250 × 81 ÷ 100 = **202.5 L**; emulsifier = 250 × 1 ÷ 100 = **2.5 (L/kg)**. Check: 45 + 202.5 + 2.5 = 250 L. ✓
2. Emulsion = 600 × 22 ÷ 100 = **132 kg**. Oil = 132 × 20 ÷ 100 = **26.4 kg**. Oil on fibre = 26.4 ÷ 600 × 100 = **4.4 %**.
3. (a) Long, strong, bright, fine tossa-type grades; roots cut, bark and specks tightly limited, no cuttings — colour uniformity is critical in an undyed warp. (b) Medium and lower grades with cuttings; shorter, duller fibre acceptable — dyeing hides colour variation and the coarse count tolerates defects.
4. The unit is a reed of ultimate fibres cemented by pectins and lignin. Dry, the cement is rigid, so pins snap the reed instead of peeling the network apart. Emulsion water plasticises the cement (given piling time) and oil lubricates, so carding separates the network progressively. Cotton's unit is already a single flexible fibre.
5. Any three: emulsion broken, so one bin sprayed unevenly; spray rate drifted (add-on too low); bin taken early, before 24–48 h conditioning completed; stricks laid too thick, so cores stayed dry; bins not rotated (unequal conditioning).